

Circuit Analysis (Lab4)

Experiment 4: Implementation of Basic Parallel Circuit

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Grade and Signature:

CLO2	Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes.			
CLO3	Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objectives: To make measurements in your own constructed parallel circuits.

Equipment required:

Power Supply, Breadboard, Connecting wires, Resistances, Multimeter.

How to make resistance measurements

Resistance is measured with the circuit's power turned off. The ohmmeter sends its own current through the unknown resistance and then measures that current to provide a resistance value readout.

Resistance measurement procedures

Follow the steps below to measure resistance:

1. Turn off the power to the circuit. Remove or isolate the component to be tested.
2. Plug the test probes into the appropriate probe jacks. Note that the jacks used may be the same ones used to measure volts.
3. Select the ohms function by turning the function switch to ohms. Start with the lowest setting.
4. Touch the probes together to check the leads, connections and battery life. The meter should display zero or a very small amount of resistance for the test leads. With the leads apart, the meter should display OL or I, depending on the manufacturer.
5. Connect the tips of the probes across the break in the component or portion of the circuit for which you want to determine resistance. If you get an OL (over limit), move to the next level.
6. View the reading on the display unit. Be sure to note the unit of measurement.
7. After all the resistance readings are complete, turn the DMM off to prevent the battery from draining

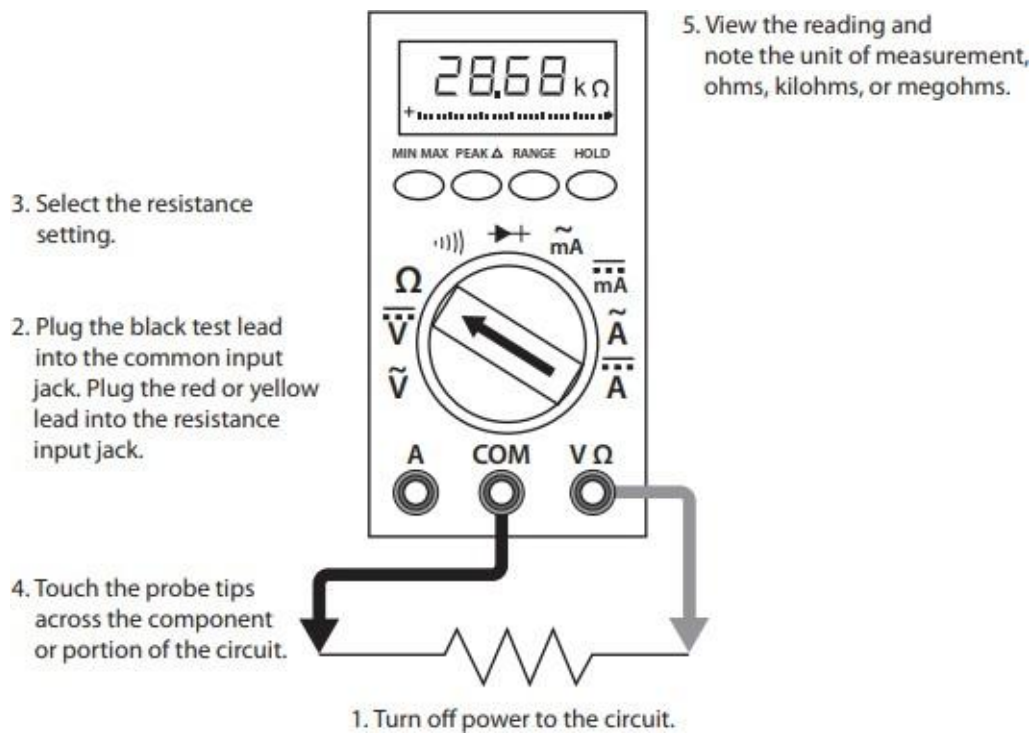


Figure 1 Using a DMM to measure resistance

How to make voltage measurements

Voltage measurements are very easy to make with a DMM. On meters with manual range selection, start with a value one setting higher than expected. An auto range DMM will automatically select the range based on the voltage present. Figure 4 shows the process.

Note:

$$1/1000 \text{ V} = 1\text{mV}$$

$$1000 \text{ V} = 1\text{kV}$$

Follow these steps to measure voltage:

1. Select the DC or AC volts function by turning the function switch to DC or AC volts.
2. Plug the test probes into the appropriate probe jacks.
3. Connect the tips of the probes across the source or load.
4. View the reading on the display unit. Be sure to note the unit of measurement. If you are testing DC voltage and a negative sign appears in the display, the polarity of your probes is incorrect and needs to be reversed.

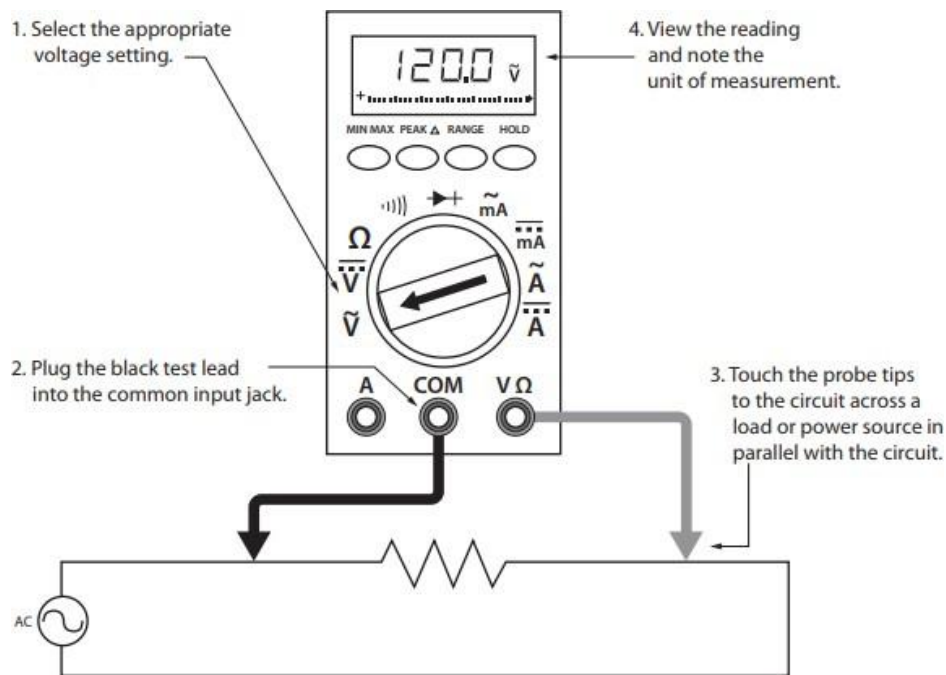


Figure 2 Using a DMM to measure voltage

How to make Current measurements

Follow the steps below to measure current:

1. Turn off the power to the circuit that will be measured.
2. Open the circuit by disconnecting or unsoldering a connection at a point where you wish to measure current.
3. Select the DC or AC amps function by turning the function switch to DC or AC amps.
4. Plug the test probes into the appropriate probe jacks. Note that the jacks used may not be the same ones used to measure volts.
5. Connect the tips of the probes across the break in the circuit so that the current to be measured flows through the meter. Note that this is a series connection. Never connect the ammeter in parallel with the source or load, as this will cause a short circuit and damage the meter.
6. Turn the circuit power back on.
7. View the reading on the display unit. Be sure to note the unit of measurement.
8. Switch power off and disconnect meter leads from the circuit.
9. If testing is finished at this test point, restore the circuit by reclosing the connection. When current measurements are complete, turn the function switch to the OFF position and remove the test leads.

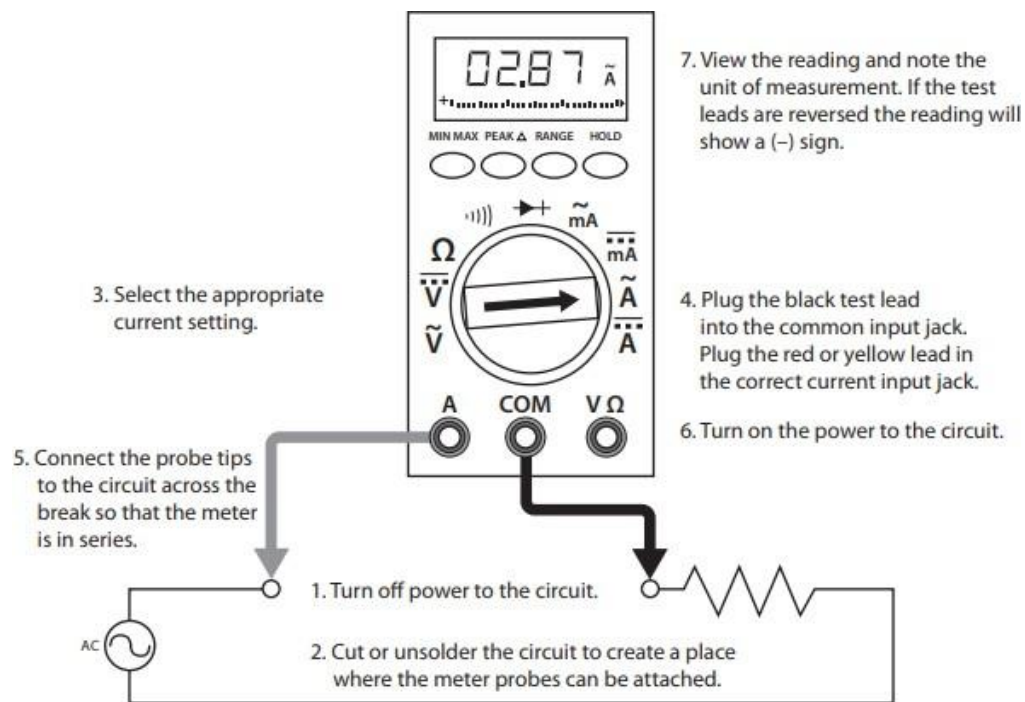
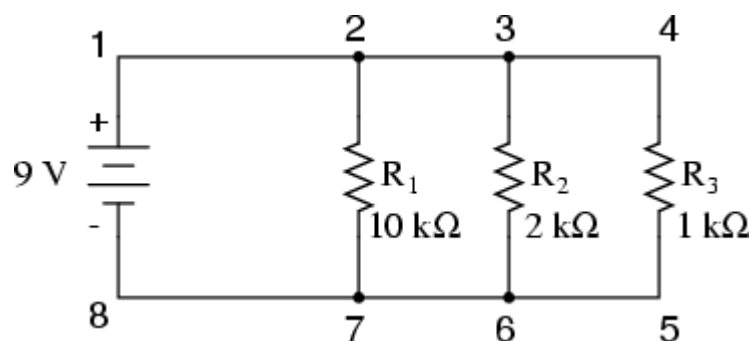


Figure 3 Using a DMM to measure current

Lab Tasks:

1. Measure the current across R1, R2, R3 and Voltage between node 2 and 7.



1. Current through R1 (10kΩ):

- $I = V / R_1$
- $I = 9V / 10k\Omega$
- **Current = 0.9 mA**

2. Current through R2 (2kΩ):

- $I = V / R_2$
- $I = 9V / 2k\Omega$
- **Current = 4.5 mA**

3. Current through R3 (1kΩ):

- $I = V / R_3$
- $I = 9V / 1k\Omega$
- **Current = 9.0 mA**

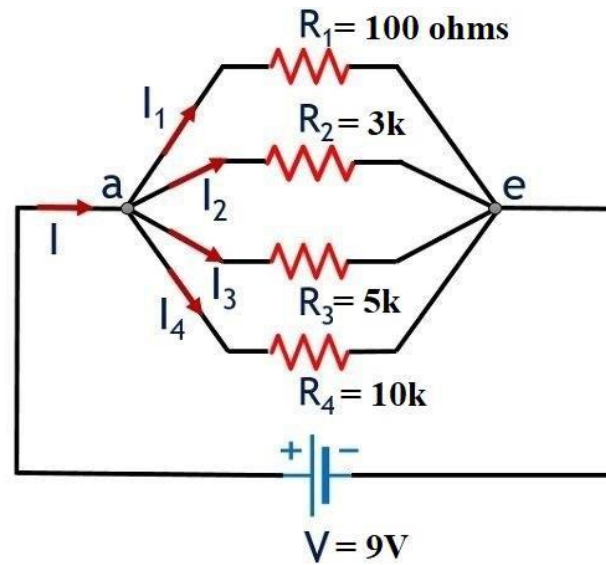
Practical Values:

R1 Current	R2 Current	R3 Current	Voltage
0.90 mA	4.50 mA	9.00 mA	9.00 V

Theoretical Values:

R1 Current	R2 Current	R3 Current	Voltage
0.9 mA	4.5 mA	9.0 mA	9.0 V

2. Measure the current across R1, R2, R3 and Voltage between node a and e.



1. Current through R₁ (100 Ω):

- $I = V / R_1$
- $I = 9V / 100 \Omega$
- $I = 0.09 \text{ A}$
- **Current = 90.0 mA**

2. Current through R₂ (3 kΩ):

- $I = V / R_2$
- $I = 9V / 3000 \Omega$
- $I = 0.003 \text{ A}$
- **Current = 3.0 mA**

3. Current through R₃ (5 kΩ):

- $I = V / R_3$
- $I = 9V / 5000 \Omega$
- $I = 0.0018 \text{ A}$
- **Current = 1.8 mA**

4. Current through R₄ (10 kΩ):

- $I = V / R_4$
- $I = 9V / 10000 \Omega$
- $I = 0.0009 \text{ A}$
- **Current = 0.9 mA**

Practical Values:

R1 Current	R2 Current	R3 Current	R4 Current	Voltage
90.0 mA	3.00 mA	1.80 mA	0.90 mA	9.00 V

Theoretical Values:

R1 Current	R2 Current	R3 Current	R4 Current	Voltage
90.0 mA	3.0 mA	1.8 mA	0.9 mA	9.0 V

Post Lab Questions

1. Why is the value of voltage across each resistor in a parallel circuit same?

In a parallel circuit, all the resistors are connected directly to the exact same two main wires (or nodes) coming from the power source. Because every branch is hooked up directly to the same starting and ending points, they all feel the exact same electrical "push" from the battery. Therefore, the voltage drop across each of them is identical.

2. When parallel resistors are of three different values, which has the greatest power loss?

The resistor with the lowest resistance value will have the greatest power loss. Because the voltage is exactly the same across all the branches, the path with the least resistance allows the most current to flow through it. Since power is calculated by multiplying voltage and current ($\text{Power} = \text{Voltage} \times \text{Current}$), the branch pulling the highest current uses up the most power.

3. How the current across any branch of a parallel circuit varies?

The current in each branch varies inversely with the resistance of that branch. According to Ohm's Law ($I = V / R$), because the voltage (V) stays constant, the current (I) depends entirely on the resistor (R). A branch with a very high resistance will "block" the electricity and only let a small amount of current through, while a branch with a very low resistance will let a large amount of current flow freely.

4. What happens to total resistance in a circuit with parallel resistors if one of them opens (i.e., removed)?

If one resistor is removed or breaks open, the total resistance of the whole circuit actually increases. You can think of parallel branches like lanes on a highway. Having many lanes makes it very easy for traffic (current) to flow, meaning the overall resistance is low. If you remove a resistor, it is like closing a lane—it becomes harder for the current to flow, which means the total resistance goes up.